

Primes and FTA



Def 1. Let $p \neq \pm 1, 0$ be an integer

(1) We say p is prime if

whenever $p \mid ab$ for $a, b \in \mathbb{Z}$.

either $p \mid a$ or $p \mid b$.

(2) We say p is irreducible if

whenever $p = ab$ for $a, b \in \mathbb{Z}$

either $a = \pm 1$ or $b = \pm 1$

Lemma 2: Let $0, \pm 1 \neq p \in \mathbb{Z}$.

Then p is prime iff p is irreducible.

proof : ($\Rightarrow$) Suppose p is prime.

We show p irreducible. Assume $p = ab$

for some $a, b \in \mathbb{Z}$. Then $p \mid ab$

implies $p \mid a$ or $p \mid b$. WLOG,

assume $p \mid a$. Write $a = np$ for $n \in \mathbb{Z}$.

Then $p = ab = npb \Rightarrow nb = 1$

which implies $n = \pm 1$, $b = \pm 1$.

($\Leftarrow$) Suppose p is irreducible.

Assume $p \mid ab$ for some $a, b \in \mathbb{Z}$

WLOG. suppose $p \nmid a$.

We claim $(p, a) = 1$.

If not. say $(p, a) = d > 1$

Then $d \mid p$ and write $p = nd$

for some $n \in \mathbb{Z}$. Since p is irreducible

and as $d > 1$, we must have $n = \pm 1$

This implies $d = p$ and $p \mid a$,

which is a contradiction to $p \nmid a$.

So the claim is proved. By Bezout's

theorem. write $1 = px + ay$ for some

$x, y \in \mathbb{Z}$. Then $b = pxb + ayb$.

Note $p \mid ab$. So $p \mid ayb$

As $p \mid pxb$. we conclude $p \mid b$

Remark 3: In $\mathbb{Z}$, irreducibles are the same as the primes is a very special property. Note Bezout's theorem is needed in the proof for ($\Leftarrow$)

Lemma 4. let $0, \pm 1 \neq n$ be an integer.

Then $n = \pm p_1 p_2 \dots p_n$ is a product of finitely many irreducibles.

proof. WLOG, assume $n > 0$

If n is irreducible, then we are done

If n is reducible, then

write $n = ab$ for $a, b > 1$

We repeat the process for a and b .

Note that $a, b < n$ and after each step we are left with smaller reducibles.

if we didn't get all irreducible factors

A decreasing sequence of positive numbers eventually terminates and thus proves finiteness.

Cor 5: let $0, \pm 1 \neq n$ be integer.

Then $n = \pm p_1 \cdots p_n$ is a finite product of primes.

Lemma 6. let $0, \pm 1 \neq n$ be integer.

Suppose $n = \pm p_1 \cdots p_s = \pm q_1 \cdots q_t$

for some $s, t \in \mathbb{Z}$. Then

$$(1) \quad s = t < \infty$$

$$(2) \quad p_i = q_i \quad \forall \quad 1 \leq i \leq s$$

$$(3) \quad p_i \leq p_j \quad \forall \quad 1 \leq i \leq j \leq s$$

proof. Suppose $p_1 \cdots p_s = q_1 \cdots q_t$

Since $p_1 \mid q_1 \cdots q_t$ and p_1 is prime.

then $p_1 \mid q_{i_1}$ for some $1 \leq i_1 \leq t$

But q_{i_1} is irreducible.

So $p_1 \mid q_{i_1} \Rightarrow q_{i_1} = p_1^{n_{i_1}} \Rightarrow n_{i_1} = \pm 1$

$$\Rightarrow p_1 = \pm q_{i_1}$$

Then consider $p_2 \cdots p_s \mid q_1 \cdots \hat{q_{i_1}} \cdots q_t$

and repeat the process for p_2 .

We see that $\exists 1 \leq i_2 \neq i_1 \leq t$ s.t

$$p_2 = \pm q_{i_2}$$

We conclude that $s \leq t$ because the map

$$\phi : \{1, 2, \dots, s\} \rightarrow \{1, 2, \dots, t\}$$

$$k \mapsto i_k$$

is injective

Conversely, we may start with

$$q_1, \dots, q_t \mid p_1, \dots, p_s$$

and conclude $t \leq s$.

So $s = t < \infty$ (n is bounded)

$$\text{Now } \phi : \{1, 2, \dots, s\} \rightarrow \{1, 2, \dots, t\}$$

being an injective map between finite sets

must be a bijection.

In particular by renaming q_i 's,

we get $p_i = \pm q_i \quad \forall 1 \leq i \leq s$

Then we may reorder everything
in nondecreasing order.

Cor T : (the fundamental theorem of Arithmetic)

let n be an integer greater than 1.

Then n has a unique prime factorization
of finite length.

Theorem 8: (Euclid)

There exists infinitely many primes.

proof. Suppose $p_1, \dots, p_n$ are
all the primes for some $n \in \mathbb{Z}$.

Consider $x = p_1 \cdots p_n + 1$.

If x is irreducible, then x is prime.

If x is reducible, then

$\exists$ a prime $p < x$ dividing x .

Then $p \in \{p_1, \dots, p_n\}$.

But $x \equiv 1 \pmod{p_i} \quad \forall 1 \leq i \leq n$.